Xinrui(Richard) Yi

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Education

Northeastern University — **San Jose, CA**
M.S. in Computer Science; ***GPA: 3.9*** — *Sep 2023 – May 2026*
- **Relevant Courses:**Machine Learning, Natural Language Processing, Distributed Systems, Cloud Computing
- **Leadership:** Career Peer Advisor, Teaching Assistant

University at Buffalo — **Buffalo, NY**
B.S. & M.S. in Civil Engineering; ***CS-Related GPA: 4.0/4.0*** — *Feb 2017 – May 2021*

Skills

Languages: Python, Java, C++, JavaScript, TypeScript, Go
AI/ML: PyTorch, TensorFlow, LangChain, LangGraph, vLLM, Claude Code, Cursor
Frameworks & Tools: Spring Boot, Django, Node.js, Flask, Flutter, Docker, Kubernetes, Git
Databases & Cloud: MySQL, MongoDB, Supabase, PostgreSQL, Redis, Kafka, AWS (S3, EC2, Lambda), GCP

Experience

Astreas Performance *(RAG, FastAPI, MLOps, Docker)* — **Mountain View, CA**
ML Engineer Intern — *Jun 2025 – Aug 2025*
- Built and scaled a production **RAG** system for a health optimization platform, processing 8,000+ daily queries from 5,000+ beta users with **99.5%** uptime on **Google Cloud Run**.
- Ran systematic module-level evaluations on the RAG pipeline via **AutoRAG** against a 100-question hand-annotated eval set, identifying retrieval as the primary latency bottleneck (2.3s avg response time) and low retrieval relevance as key improvement.
- Implemented **Weaviate** with **HNSW** indexing and hybrid search (dense + sparse retrieval with **RRF** fusion), combined with overlapping chunking and a **cross-encoder** reranker over top-20 candidates, cutting retrieval latency by ∼60% and improving relevance from 0.73 to 0.89 on **NDCG@10**.
- Integrated **PathRAG** (graph-based RAG) over an LLM-constructed health knowledge graph, enabling multi-hop retrieval across nutrition, training, and sleep domains; improved retrieval **recall** from 72% to 90%.
- Deployed a containerized RAG service (**FastAPI** + **Streamlit**) on GCP Cloud Run with **Docker**, handling ∼200 peak concurrent users during beta.

Voc.ai *(AI Agents, Prompt Engineering, Web Crawling, Client Demos)* — **San Jose, CA**
AI Agent Engineer Intern — *Jan 2025 – Apr 2025*
- Built customer-facing **AI agent workflows** on the company's internal agent platform, composing built-in tools (**database query**, **web crawler**) into end-to-end flows that answered e-commerce shopper questions such as order and shipment status.
- Integrated client-provided order databases and crawled product/pricing pages into agent **knowledge bases**, grounding responses in each client's own catalog and order data.
- Prepared and ran ∼10-query demo scenarios covering common support intents, presenting live agent behavior to enterprise clients during pre-sales evaluations.
- Used the platform's feedback dashboard to review end-user-flagged unsatisfactory responses, diagnosing whether each failure came from **intent misrouting** or **response generation**, and iterating accordingly by adding intents to the generation prompt.

Selected Projects

Agent Teams: Multi-Agent Orchestration Platform | ***LangGraph, Deep Agents, Next.js, TypeScript*** **(Github Link)**
- Built at a hackathon: a Slack-style app where each channel is a team of AI agents (**Next.js**/**React** frontend, **Node.js** backend).
- Orchestrated a CEO → PM → specialist **multi-agent** hierarchy with **LangChain Deep Agents** on **LangGraph**: the CEO staffs a team from a goal, the PM assigns tasks, and specialists call tools (web search, page fetch) and report back.
- Ran persistent agents with task queues, inboxes and **SSE** streaming; stored agent memory in a **git-backed vault** with diff and restore.
- Routed LLM calls with automatic **model fallback**; logged cost, tokens, cache hits and latency per call with **LangSmith**.

Rissa Recovery: Compliance-Aware Debt Collection Voice Agent | ***LiveKit, LangGraph, TTS*** **(Github Link)**
- Built a real-time debt collection voice agent supporting identity verification, objection handling, payment-plan negotiation, recap, and call closing through a browser-based call interface.
- Designed a **LangGraph** dialogue state machine with per-turn routing and persistent call state, publishing live transcript, stage, emotion, and verification signals to the **React** frontend.
- Integrated **Deepgram** speech-to-text, **Gemini** response generation, **LiveKit** WebRTC transport, and **Fish Audio TTS** to enable low-latency end-to-end voice interaction.
- Implemented an **FDCPA compliance gate** using regex checks and LLM-based fallback classification to audit generated responses before speech synthesis.And authored supporting **project documentation** outlining the end-to-end development process.